

4. Arrays

1. Write a function to add integer values of an array

Answer:

```
public class AddArrayValues {

    // Function to add array elements
    public static int sumArray(int[] arr) {
        int sum = 0;

        // Loop through array
        for (int i = 0; i < arr.length; i++) {
            sum = sum + arr[i];
        }

        return sum;
    }

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        // Call function
        int result = sumArray(numbers);

        // Print result
        System.out.println("Sum of array elements: " + result);
    }
}
```

2. Write a function to calculate the average value of an array of integers

Answer:

```
public class AverageArray {

    // Function to calculate average
    public static double findAverage(int[] arr) {

        int sum = 0;

        // Add all elements
        for (int i = 0; i < arr.length; i++) {
            sum = sum + arr[i];
        }

        // Calculate average
        double average = (double) sum / arr.length;

        return average;
    }

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};
```

```

        // Call function
        double avg = findAverage(numbers);

        // Print average
        System.out.println("Average value: " + avg);
    }
}

```

3. Write a program to find the index of an array element

Answer:

```

public class FindIndex {

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        int searchElement = 30;
        int index = -1;

        // Find index
        for (int i = 0; i < numbers.length; i++) {

            if (numbers[i] == searchElement) {
                index = i;
                break;
            }
        }

        // Print result
        if (index != -1) {
            System.out.println("Element found at index: " + index);
        } else {
            System.out.println("Element not found");
        }
    }
}

```

4. Write a function to test if array contains a specific value

Answer:

```

public class ContainsValue {

    // Function to check value
    public static boolean contains(int[] arr, int value) {

        for (int i = 0; i < arr.length; i++) {

            if (arr[i] == value) {
                return true;
            }
        }

        return false;
    }
}

```

```

    }

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        boolean result = contains(numbers, 40);

        System.out.println("Contains value? " + result);
    }
}

```

5. Write a function to remove a specific element from an array

Answer:

```

import java.util.Arrays;

public class RemoveElement {

    // Function to remove element
    public static int[] removeElement(int[] arr, int element) {

        int[] newArray = new int[arr.length - 1];
        int index = 0;

        for (int i = 0; i < arr.length; i++) {

            if (arr[i] != element) {
                newArray[index] = arr[i];
                index++;
            }
        }

        return newArray;
    }

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        int[] result = removeElement(numbers, 30);

        System.out.println(Arrays.toString(result));
    }
}

```

6. Write a function to copy an array to another array

Answer:

```

import java.util.Arrays;

public class CopyArray {

    public static void main(String[] args) {

```

```

int[] original = {10, 20, 30, 40, 50};

// Create new array
int[] copied = new int[original.length];

// Copy elements
for (int i = 0; i < original.length; i++) {
    copied[i] = original[i];
}

// Print copied array
System.out.println(Arrays.toString(copied));
}
}

```

7. Write a function to insert an element at a specific position in the array

Answer:

```

import java.util.Arrays;

public class InsertElement {

    public static int[] insertElement(int[] arr, int element, int position)
    {

        int[] newArray = new int[arr.length + 1];

        // Copy before position
        for (int i = 0; i < position; i++) {
            newArray[i] = arr[i];
        }

        // Insert element
        newArray[position] = element;

        // Copy remaining elements
        for (int i = position; i < arr.length; i++) {
            newArray[i + 1] = arr[i];
        }

        return newArray;
    }

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40};

        int[] result = insertElement(numbers, 25, 2);

        System.out.println(Arrays.toString(result));
    }
}

```

8. Write a function to find the minimum and maximum value of an array

Answer:

```
public class MinMaxArray {

    public static void main(String[] args) {

        int[] numbers = {50, 10, 90, 30, 70};

        int min = numbers[0];
        int max = numbers[0];

        // Find min and max
        for (int i = 1; i < numbers.length; i++) {

            if (numbers[i] < min) {
                min = numbers[i];
            }

            if (numbers[i] > max) {
                max = numbers[i];
            }
        }

        System.out.println("Minimum value: " + min);
        System.out.println("Maximum value: " + max);
    }
}
```

9. Write a function to reverse an array of integer values

Answer:

```
import java.util.Arrays;

public class ReverseArray {

    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        // Reverse array
        for (int i = 0; i < numbers.length / 2; i++) {

            int temp = numbers[i];
            numbers[i] = numbers[numbers.length - 1 - i];
            numbers[numbers.length - 1 - i] = temp;
        }

        System.out.println(Arrays.toString(numbers));
    }
}
```

10. Write a function to find the duplicate values of an array

Answer:

```
public class DuplicateValues {
```

```

public static void main(String[] args) {

    int[] numbers = {10, 20, 30, 20, 40, 10, 50};

    System.out.println("Duplicate values are:");

    // Compare each element
    for (int i = 0; i < numbers.length; i++) {

        for (int j = i + 1; j < numbers.length; j++) {

            if (numbers[i] == numbers[j]) {
                System.out.println(numbers[i]);
                break;
            }

        }

    }

}

```

11. Write a program to find the common values between two arrays

Answer:

```

public class CommonValues {

    public static void main(String[] args) {

        int[] array1 = {10, 20, 30, 40, 50};
        int[] array2 = {30, 40, 60, 70};

        System.out.println("Common values are:");

        // Compare arrays
        for (int i = 0; i < array1.length; i++) {

            for (int j = 0; j < array2.length; j++) {

                if (array1[i] == array2[j]) {
                    System.out.println(array1[i]);
                }

            }

        }

    }

}

```

12. Write a method to remove duplicate elements from an array

Answer:

```

import java.util.Arrays;

public class RemoveDuplicates {

    public static void main(String[] args) {

```

```

int[] numbers = {10, 20, 20, 30, 40, 40, 50};

int[] temp = new int[numbers.length];
int count = 0;

// Remove duplicates
for (int i = 0; i < numbers.length; i++) {

    boolean isDuplicate = false;

    for (int j = 0; j < count; j++) {

        if (numbers[i] == temp[j]) {
            isDuplicate = true;
            break;
        }
    }

    if (!isDuplicate) {
        temp[count] = numbers[i];
        count++;
    }
}

System.out.println("Array after removing duplicates:");

for (int i = 0; i < count; i++) {
    System.out.print(temp[i] + " ");
}
}

```

13. Write a method to find the second largest number in an array

Answer:

```

public class SecondLargest {

    public static void main(String[] args) {

        int[] numbers = {10, 80, 50, 30, 90, 70};

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        // Find second largest
        for (int i = 0; i < numbers.length; i++) {

            if (numbers[i] > largest) {

                secondLargest = largest;
                largest = numbers[i];

            } else if (numbers[i] > secondLargest
                && numbers[i] != largest) {

```

```

        secondLargest = numbers[i];
    }
}

System.out.println("Second largest number: " + secondLargest);
}
}

```

14. Write a method to find the second largest number in an array

Answer:

```

public class SecondLargest {

    public static void main(String[] args) {

        int[] numbers = {15, 25, 35, 45, 55};

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for (int i = 0; i < numbers.length; i++) {

            if (numbers[i] > largest) {

                secondLargest = largest;
                largest = numbers[i];

            } else if (numbers[i] > secondLargest
                && numbers[i] != largest) {

                secondLargest = numbers[i];
            }
        }

        System.out.println("Second largest number: " + secondLargest);
    }
}

```

15. Write a method to find number of even number and odd numbers in an array

Answer:

```

public class EvenOddCount {

    public static void main(String[] args) {

        int[] numbers = {10, 15, 20, 25, 30, 35};

        int evenCount = 0;
        int oddCount = 0;

        // Count even and odd
        for (int i = 0; i < numbers.length; i++) {

            if (numbers[i] % 2 == 0) {

```



```

        evenCount++;
    } else {
        oddCount++;
    }
}

System.out.println("Even numbers count: " + evenCount);
System.out.println("Odd numbers count: " + oddCount);
}
}

```

16. Write a function to get the difference of largest and smallest value

Answer:

```

public class DifferenceLargestSmallest {

    public static void main(String[] args) {

        int[] numbers = {10, 50, 90, 20, 70};

        int min = numbers[0];
        int max = numbers[0];

        // Find min and max
        for (int i = 1; i < numbers.length; i++) {

            if (numbers[i] < min) {
                min = numbers[i];
            }

            if (numbers[i] > max) {
                max = numbers[i];
            }
        }

        int difference = max - min;

        System.out.println("Difference: " + difference);
    }
}

```

17. Write a method to verify if the array contains two specified elements(12,23)

Answer:

```

public class ContainsTwoElements {

    public static void main(String[] args) {

        int[] numbers = {10, 12, 20, 23, 50};

        boolean has12 = false;
        boolean has23 = false;

        // Check array
        for (int i = 0; i < numbers.length; i++) {

```

```

        if (numbers[i] == 12) {
            has12 = true;
        }

        if (numbers[i] == 23) {
            has23 = true;
        }
    }

    // Print result
    if (has12 && has23) {
        System.out.println("Array contains both 12 and 23");
    } else {
        System.out.println("Array does not contain both elements");
    }
}
}

```

18. Write a program to remove the duplicate elements and return the new array

Answer:

```

import java.util.Arrays;

public class NewArrayWithoutDuplicates {

    public static void main(String[] args) {

        int[] numbers = {10, 20, 20, 30, 40, 40, 50};

        int[] temp = new int[numbers.length];
        int count = 0;

        // Remove duplicates
        for (int i = 0; i < numbers.length; i++) {

            boolean duplicate = false;

            for (int j = 0; j < count; j++) {

                if (numbers[i] == temp[j]) {
                    duplicate = true;
                    break;
                }
            }

            if (!duplicate) {
                temp[count] = numbers[i];
                count++;
            }
        }

        // Create new array
        int[] newArray = new int[count];
    }
}

```

```

        for (int i = 0; i < count; i++) {
            newArray[i] = temp[i];
        }

        System.out.println(Arrays.toString(newArray));
    }
}

```

19. Write a function to find the missing number of sorted array of 1 to 100.

Answer:

```

public class MissingNumber {

    public static void main(String[] args) {

        // Example array with missing number 55
        int[] numbers = new int[99];

        int index = 0;

        // Create array from 1 to 100 except 55
        for (int i = 1; i <= 100; i++) {

            if (i != 55) {
                numbers[index] = i;
                index++;
            }
        }

        int expectedSum = 100 * 101 / 2;
        int actualSum = 0;

        // Find actual sum
        for (int i = 0; i < numbers.length; i++) {
            actualSum = actualSum + numbers[i];
        }

        int missingNumber = expectedSum - actualSum;

        System.out.println("Missing number: " + missingNumber);
    }
}

```